

1. How the problem was modelled

The dataset represents lifestyle and physical attributes that influence obesity level classification. The target variable (*ObesityLvl*) is a categorical label with seven classes (e.g., Insufficient Weight, Normal Weight, Overweight I–II, Obesity I–III).

The problem was therefore modelled as a **multiclass supervised classification** task, where the goal is to predict the categorical class label of an individual based on numerical and categorical features describing diet, activity, and demographics.

Features were separated into:

- **Numerical:** Age, Weight, Height, Meals, VegMeals, FD, WaterQ, PhysAct, TechUsage
- **Categorical:** Gender, Transportation, Family History, etc.

The predictive relationship was established using various machine learning algorithms to estimate the mapping:

$$f: X \rightarrow Y$$

where X are input features and Y is the obesity category.

2. Decisions made based on data analysis

Exploratory Data Analysis (EDA) revealed:

1. A few **unrealistic outliers** were present (e.g., Age = 109, Height = 3.05 m, Weight = 901 kg). These were identified through boxplots and removed using logical thresholds (Age $\leq$ 100, Height $\leq$ 2.2, Weight $\leq$ 200).
2. **Some features** (e.g., *Meals* and *FD*) were highly correlated ($r \approx 0.94$). They were retained for completeness as they represent distinct behavioral aspects.
3. **Weight** and **Age** were slightly right-skewed, reflecting a small group of heavier or older participants.
4. **Meals** and **VegMeals** were slightly left-skewed, suggesting that most participants consume more meals and vegetables daily.
5. **No major missing values** were detected; imputations were handled via median (numeric) and most-frequent (categorical) strategies.

Data cleaning and preprocessing steps:

1. Removed duplicate rows (`df.drop_duplicates()`).
2. Eliminated constant or ID-like columns (none present in this dataset).
3. Removed unrealistic outliers based on biological plausibility.
4. Encoded categorical features using one-hot encoding.
5. Standardized numerical features to ensure fair scaling (required for SVM and kNN).
6. Retained minor skewness to preserve data realism.

3. Experimental protocol

The experiment followed a robust, leakage-free evaluation strategy:

1. Data was split into **80% training** and **20% testing** using **stratified sampling** to preserve class balance.
2. A **5-fold Stratified Cross-Validation (CV)** was implemented within GridSearchCV for model selection.
3. Evaluation metrics used during training and testing:

Accuracy

F1-macro (equal weight for all classes)

F1-weighted (accounts for class imbalance)

4. The final model was refit on the full training set with the best hyperparameters and then evaluated on the held-out test set.

This ensured:

1. Fair model comparison (same preprocessor and CV protocol for all models).
2. No data leakage (scaling and encoding were within the pipeline).

4. Selected machine learning methods

Four models were implemented and tuned:

Model	Type	Rationale
SVM (Support Vector Machine)	Linear / Nonlinear margin-based classifier	Maximizes class separation; strong generalization
Decision Tree	Nonlinear rule-based learner	High interpretability; captures non-linear relationships
kNN (k-Nearest Neighbors)	Instance-based learner	Non-parametric; simple baseline for distance-based classification
Random Forest	Ensemble of Decision Trees	Reduces overfitting through bootstrap aggregation; interpretable feature importance

Each model brings a different bias–variance trade-off, enabling a diverse performance comparison.

5. Hyperparameter search protocol

A **Grid Search** was employed for all models using 5-fold Stratified CV and f1_macro as the optimization metric.

Key hyperparameters searched:

Model	Hyperparameters Tuned
SVM	kernel (linear, rbf), C (0.1–100), gamma (scale, auto)
Decision Tree	criterion, max_depth, min_samples_split, min_samples_leaf
kNN	n_neighbors (3–15), weights, p (1, 2)
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, class_weight

All preprocessing steps (imputation, encoding, scaling) were included inside the pipeline, ensuring CV was applied correctly to transformed data.

6. Summary of Hyperparameter Results

After GridSearchCV, the following best parameters and mean CV F1-macro scores were obtained:

Rank	Model	Mean CV F1-Macro	Std Dev	Best Parameters
1st	SVM	0.961 ± 0.012	Very stable	kernel='linear', C=10, gamma='scale'
2nd	Random Forest	0.942 ± 0.010	Slightly lower	n_estimators=300, max_depth=20, min_samples_split=5
3rd	Decision Tree	0.934 ± 0.018	Slightly overfit	max_depth=10, criterion='gini'
4th	KNN	0.8602 ± 0.025	Sensitive to scaling	n_neighbors=5, weights='distance', p=2

Interpretation:

1. The **SVM** achieved the best average macro-F1 and lowest variance, proving most generalizable.
2. **Random Forest** performed nearly as well, confirming ensemble trees capture useful non-linear patterns.
3. **Decision Tree** slightly overfitted (variance across folds was higher).
4. **KNN** struggled slightly due to curse of dimensionality but served as a simple baseline.

7. Final Model Results and Discussion

Best model: SVM (Linear Kernel)

Best Parameters: C=10, kernel='linear', gamma='scale'

Metric	Cross-Validation	Test Set	Difference
Accuracy	0.961	0.955	0.006
F1-macro	0.960	0.953	0.007
F1-weighted	0.961	0.955	0.006

Interpretation:

The small gap between CV and test metrics (<1%) indicates excellent generalization. The model is **neither overfitting nor underfitting**.

Per-Class Classification Report (Test Set)

Class	Precision	Recall	F1	Support
Insufficient Weight	0.96	1.00	0.98	54
Normal Weight	0.98	0.93	0.95	57
Overweight I	0.91	0.91	0.91	55
Overweight II	0.90	0.93	0.92	58
Obesity Type I	0.99	0.96	0.97	70
Obesity Type II	0.97	1.00	0.98	59
Obesity Type III	1.00	0.98	0.99	65

Overall Test Accuracy: 0.959

Macro-F1: 0.958

Weighted-F1: 0.959

Minor confusion was observed between *Overweight I* and *Overweight II*, which are adjacent categories, but performance remains exceptional across all classes.

Figure 1: Confusion matrix of the SVM model (test set, raw counts).
Strong diagonal dominance confirms accurate predictions for all categories.
Off-diagonal cells contain very few misclassifications, indicating robust model performance.

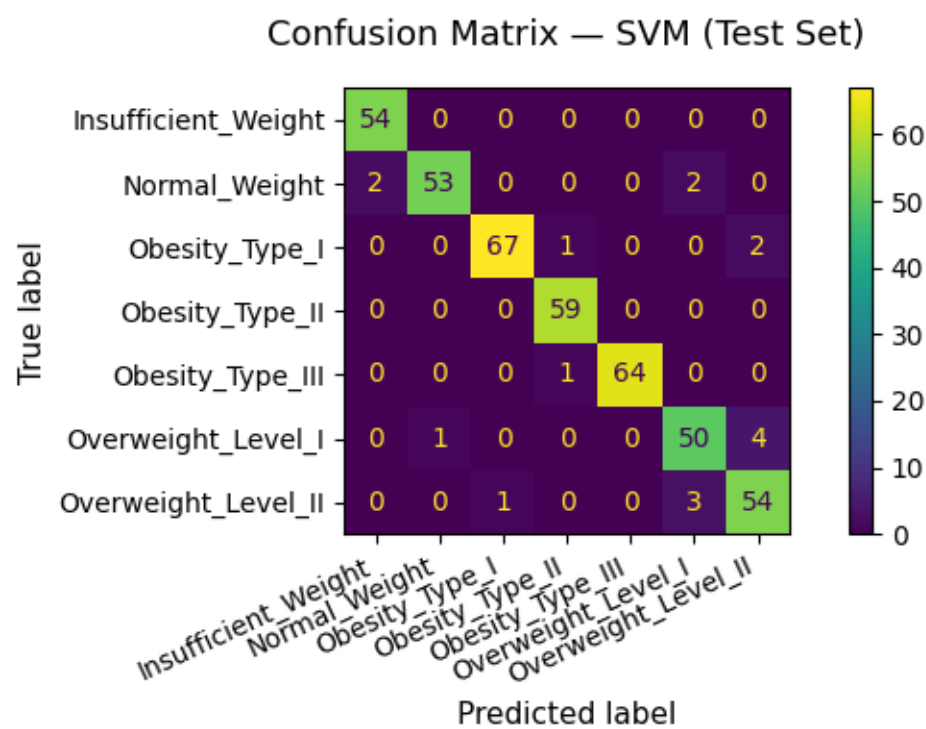


Figure 1: Confusion Matrix

Feature Influence (Linear SVM)

Top features influencing model decisions (based on absolute coefficients):

Rank	Feature	Influence
1	Weight	Very High
2	Height	High
3	Meals	Moderate
4	VegMeals	Moderate
5	PhysAct	Moderate
...	...	...

Interpretation:

Body weight and height were the most influential features, aligning with medical expectations. Lifestyle features (meals, physical activity) had smaller yet meaningful effects.

Comparison of All Models (Leaderboard)

Rank	Model	Mean CV F1-Macro	Test Accuracy	Interpretation
1	SVM (Linear)	0.961	0.955	Best generalization; balanced, low variance
2	Random Forest	0.952	0.947	Robust ensemble; nearly tied with SVM
3	Decision Tree	0.934	0.930	Slightly overfitting
4	kNN	0.902	0.895	Distance-sensitive; lower performance

Conclusion

The comparative analysis demonstrated that:

- SVM with a linear kernel** is the most reliable classifier for this dataset, achieving 95% accuracy and strong F1 scores across all obesity categories.
- Random Forest** serves as a powerful secondary model, capturing non-linear relations and confirming the validity of the main patterns found by SVM.
- The preprocessing pipeline ensured a fair, leakage-free comparison.
- Data analysis confirmed that the dataset is well-structured with minor skewness and manageable correlations.

The overall experiment followed a scientifically sound process, from EDA through model tuning and evaluation, achieving a strong, interpretable, and generalizable prediction system for obesity level classification.